Clinical Care-Plan Case Study: Hip Fracture in a Frail Patient with Multiple Chronic Conditions

ADN Med-Surg II | Individual take-home clinical care-plan case study | Open book

Your patient

You are the primary nurse for the patient in Step-Down ICU room 378-1 on the 7:00 to 15:00 shift on Thursday, October 8, 2026 (hospital day 1). Open the patient in the student EHR (Student View, or the patient file from your instructor) and set the simulation clock to 0700. Everything you need is in the chart. She is waiting for surgery to repair a broken hip, and she has many other health problems. Work the way you would at the start of a real shift.

How this case study works

This packet follows the nursing process: Assessment, Diagnosis, Planning, Implementation and Evaluation. You will also explain the pathophysiology and pharmacology behind your patient's care. Expect to spend about 6 hours. It is a take-home, individual, open-book assignment: use your textbooks, drug guide, notes and the chart.

Questions ask you to explain WHY, not only WHAT. Short, specific answers that use this patient's data earn more points than general textbook answers. Real charts are imperfect: you will be asked to catch conflicts and to question orders before you act.

Part	Nursing process step	Estimated time
1	Assessment: Patient Data and Chart Review (Assessment)	80 min
2	Pathophysiology: The Disease Process (Assessment / Analysis of cues)	75 min
3	Pharmacology and Medication Safety (Assessment / Analysis (and safe implementation))	70 min
4	Nursing Diagnoses and Priorities (Diagnosis)	30 min
5	Planning and Implementation (Planning / Implementation)	60 min
6	Evaluation: What Happened and What Next? (Evaluation)	35 min
7	Clinical Judgment (Next-Generation NCLEX Style) (Evaluation / Clinical judgment)	30 min
	Estimated total	**6 h 20 min**

Scoring

- The packet is worth 100 points. A score of 78 or higher is passing.
- Each numbered item shows its point value. Points are given only for what is written in the box.

Writing so your pages can be scored from photos

- Write in dark pen or very dark pencil, print clearly, and stay inside the box. Write the answer in the box with the same number as the question.
- Write your name and date at the top of every page (the header prints a line for it).
- If you need more room, write "continued on the back" and put the item number on the back. Do not squeeze writing between boxes.
- Do not use abbreviations that are not on the approved list. Spell out drug names.
- When you finish: photograph EVERY page, one page per photo, on a flat surface in good light with all four corners and the page number in view. Check that every photo is readable before you upload.

Part 6 is sealed

- Do not read the sealed page at the start of Part 6 until Parts 1 to 5 are finished. Your plan must be based on what you knew at 0700.

Honor statement: I completed this work myself, using only my textbooks, notes, drug references and the patient chart.

Signature: _____ Date: _____

PART 1: ASSESSMENT: PATIENT DATA AND CHART REVIEW

Nursing process step: Assessment | About 80 minutes | 26 points

Collect and organize the data you would have at 0700. Where the chart does not document something, write "not documented." Use R and L for laterality.

1.1 Basic Patient Information *(4 pts)*

Where to look in the EHR: Patient Summary and Storyboard

Room Number:		**Age:**	
Gender:		**Admission Date:**	
Allergies (and reactions):		**Chief Complaint:**	
Primary Diagnosis:		**Code Status:**	
Isolation:			

1.2 Medical Diagnoses and How They Affect the Primary Diagnosis *(4 pts)*

List FIVE other medical diagnoses from this patient's chart. For each, explain how it affects her hip fracture and her surgery.

Where to look in the EHR: Chart Review > H&P; Encounters > Hospital problem list

Medical diagnosis	How it affects the primary diagnosis (hip fracture and its surgery)

1.3 Patient History *(3 pts)*

Where to look in the EHR: H&P; ED Provider Note; Case Management consult order; Nursing Admission Note

Medical History:

Surgical History (with years):

Family History:

Social History:

Home Situation and Discharge Needs:

Pre-Hospital Story of Patient:

1.4 Head-to-Toe Assessment at 0700 (use R and L for laterality) *(4 pts)*

Document the findings from the chart as of the start of your shift. If the chart does not document a finding, write "not documented."

Where to look in the EHR: Flowsheets; Nursing Admission Note 0536; General Surgery H&P

Neurological:

Cardiovascular: Radial pulses R___ L___ Pedal pulses R___ L___ Cap refill _____

Respiratory: RUL ____ LUL ____ RLL ____ LLL ____ (breath sounds, effort, O2)

Integumentary:

Musculoskeletal:

Gastrointestinal:

Genitourinary:

IV / Lines / Devices Assessment:

Psychosocial / Cultural Assessment:

Erikson Stage:

1.5 Vital Sign Trend *(2 pts)*

Copy the vital signs at each time. Circle every abnormal value. Then describe the trend in one sentence.

Where to look in the EHR: Flowsheets > Vital Signs (or the Vitals graph)

Vital sign	10/08 0206 (ED arrival)	10/08 0300	10/08 0424
Temperature			
Heart rate			
Blood pressure			
Respiratory rate			
SpO2 (%)			
Pain (0-10)			

Trend in one sentence:

1.6 Laboratory Data *(4 pts)*

Choose the 8 laboratory results most important for this patient before surgery. For each give the current value, the normal range (this is her first set, so there is no trend yet), and explain why it matters for THIS patient.

Where to look in the EHR: Lab Results

Lab	Result / normal range	Why is this lab important for this patient?

1.7 Diagnostics *(2 pts)*

Where to look in the EHR: Chart Review > Imaging, Cardiology

X-Ray / CT / MRI: location and results (include the chest)

Echocardiogram results:

EKG results:

Urinalysis results:

1.8 Chart Reconciliation *(3 pts)*

Real charts contain conflicts and gaps. Find FOUR places where this chart is conflicting, duplicated, or needs clarification before you act. Say where you found each and who you would clarify with.

Where to look in the EHR: Orders, MAR, Chart Review > Notes, Sticky notes

What is wrong or unclear (and where)	Who you would clarify with and what you would ask

PART 2: PATHOPHYSIOLOGY: THE DISEASE PROCESS

Nursing process step: Assessment / Analysis of cues | About 75 minutes | 20 points

Explain WHY this is happening. Use your own words (plain language) and tie each idea to this patient.

2.1 Pathophysiology of an Intertrochanteric Hip Fracture *(6 pts)*

In plain language and your own words, explain the pathophysiology of this patient's primary diagnosis: why the bone broke from a fall from standing height, why her leg looks the way it does, what the body does in response, and why the problem is treated with surgery quickly.

2.2 Cause-and-Effect Chain *(3 pts)*

Fill in each box (one or two lines) to show how events in this patient lead from the fall to the problems you will have to prevent.

1. Why did she fall? (factors in THIS patient)	

▼

2. Why did the bone break so easily?	

▼

3. What is broken and what does the leg look like?	

▼

4. What is the body's immediate response (pain, stress, bleeding)?	

▼

5. What happens because she cannot move?	

▼

6. How does the nail surgery and early mobility interrupt the chain?	

2.3 How Do You Know? Findings and Mechanisms *(4 pts)*

List EIGHT findings from this patient's assessment, labs or diagnostics that support the hip fracture or show how her body is responding. Beside each, explain the mechanism that causes it.

Where to look in the EHR: Notes, Flowsheets, Lab Results, Imaging

Finding in THIS patient (with value)	Why it happens (mechanism)

2.4a Why her bones are fragile *(1 pt)*

Name THREE things in her chart (medicines, diagnoses, habits or labs) that weakened her bones and explain how each does it.

2.4b Heart and lungs around surgery *(1 pt)*

She has HFrEF (EF about 35%), coronary artery disease, atrial fibrillation and COPD. Explain TWO ways surgery and blood loss threaten her heart and lungs, and what that means for fluids, beta blockers and oxygen.

2.4c Steroids, methotrexate and infection *(1 pt)*

She takes prednisone 7.5 mg daily and methotrexate weekly for rheumatoid arthritis and has a history of MRSA. Explain how these affect her risk for infection and how an infection might look different in her.

2.5 Treatment *(2 pts)*

What are we doing to treat this diagnosis before and after surgery? Include medical treatments AND nursing monitoring.

2.6 Hemoglobin *(2 pts)*

Her hemoglobin is 8.9 (baseline about 10). Why does this matter more in HER than in a healthy patient having the same surgery? State the transfusion threshold that is ordered, what conflicting threshold you found in the notes, and what you would watch for if she needs blood.

Where to look in the EHR: Lab Results > Hemoglobin; Orders

PART 3: PHARMACOLOGY AND MEDICATION SAFETY

Nursing process step: Assessment / Analysis (and safe implementation) | About 70 minutes | 14 points

Use your drug guide and the MAR (and the drug guide link inside the EHR). Many home medications were ordered at admission; part of your job is to question them.

3.1 Medication List *(6 pts)*

From the MAR and orders, choose the EIGHT medications that matter most to you before the 0900 medication pass. List brand and generic names.

Where to look in the EHR: MAR (click the medication name for the drug guide)

Medication (brand and generic)	Drug class and mechanism (10 words or fewer)	Why is THIS patient taking it?	Key nursing check / hold parameter

3.2a Blood thinners around surgery *(1.5 pts)*

Look at the heparin, enoxaparin, apixaban and aspirin orders and their start dates. Explain the plan the orders are following before and after surgery, what you must verify with the patient and the team before the OR, what you assess, and why her bleeding risk is higher than average.

Where to look in the EHR: MAR, Orders (note the start dates), Sticky notes

Write one answer per numbered box, inside the box, in dark pen, neatly. Open book, individual work.

3.2b Beta blockers, ACE inhibitor and diuretic *(1.5 pts)*

At 0900 she has metoprolol AND carvedilol, lisinopril, and furosemide due. What is wrong or risky here, what do you do before giving any of them, and what do the hold parameters protect against?

3.2c Methotrexate and prednisone *(1.5 pts)*

Methotrexate 15 mg is a weekly high-alert medicine and prednisone is a long-term steroid. What must you verify about each before surgery, and what could go wrong?

3.2d Insulin and glucose *(1.5 pts)*

Her glucose is 201 and A1c 8.0. She has glargine 20 units at bedtime and a lispro correction scale, will be NPO after midnight, and takes prednisone. What is the glucose goal, what are the risks, and what will you monitor and do?

3.3 Math From the Chart *(2 pts)*

Show your work in the box. Include units.

Where to look in the EHR: MAR, Orders, Nerve block procedure note

a) The fascia iliaca block used 30 mL of ropivacaine 0.2%. How many mg were given? The maximum is 3 mg/kg; is this below the maximum for her 70 kg?

b) Cefazolin 2 g is mixed in 100 mL and ordered to infuse over 30 minutes. What is the pump rate in mL/hour?

c) Her glargine dose of 20 units is a 20% reduction from her home dose. What is her home dose?

d) Maintenance lactated Ringer's runs at 40 mL/hour. How many mL will she receive in 3 hours? (She has HFrEF; why does the volume matter?)

PART 4: NURSING DIAGNOSES AND PRIORITIES

Nursing process step: Diagnosis | About 30 minutes | 8 points

Use NANDA-I diagnoses. Write each as problem, "related to" (cause you can treat), and "as evidenced by" (specific data). Do not use a medical diagnosis as the "related to."

4.1 Prioritized Nursing Diagnoses *(6 pts)*

Write FOUR nursing diagnoses in priority order (1 = highest).

#	NANDA-I nursing diagnosis	Related to (etiology)	As evidenced by (at least 2 specific chart findings)

4.2 Why This Order? *(2 pts)*

Explain why your #1 diagnosis is first (use ABCs, Maslow, or acuity). Name one risk diagnosis you ranked lower that could become the top priority and what finding would move it up.

PART 5: PLANNING AND IMPLEMENTATION

Nursing process step: Planning / Implementation | About 60 minutes | 14 points

Plan the care for the 0700-1500 shift. She is waiting for surgery. SMART goal = Specific, Measurable, Achievable, Relevant, Time-bound, written about the PATIENT (not the nurse).

5.1a Plan for Priority Diagnosis #1: _____ *(2.5 pts)*

I = independent, C = collaborative, D = delegated.

SMART goal for this shift	Nursing interventions (at least 4: what, how often)	Rationale	I / C / D

5.1b Plan for Priority Diagnosis #2: _____ *(2.5 pts)*

SMART goal for this shift	Nursing interventions (at least 4: what, how often)	Rationale	I / C / D

5.2 Time-Sequenced Plan for the Shift *(3 pts)*

For each hour, list your priority actions IN ORDER. Include medications (and which you will question), pre-operative preparation, assessments, teaching, and communication, and say why that time.

Where to look in the EHR: Brain/Worklist, MAR, Orders, Secure Chat, Notes

Time	Priority actions in order	Why at this time
0700		
0800		
0900		
1000		
1100		
1200		
1300		
1400		

5.3 Calling the Provider (SBAR) *(2 pts)*

At 0830 you review the 0900 medications. Metoprolol tartrate AND carvedilol are both scheduled, lisinopril and furosemide are due although the hospitalist said to hold diuretics, her BP is 150/89 and HR 99, and the OR time is unclear. You decide to call the hospitalist (co-management). Write what you would say.

Where to look in the EHR: MAR; Orders; Notes

S: Situation

Write one answer per numbered box, inside the box, in dark pen, neatly. Open book, individual work.

B: Background

A: Assessment

R: Recommendation / request

5.4 Delegation *(2 pts)*

Mark the LOWEST level of staff who may be assigned each task for this patient.

	RN only	LPN/LVN	UAP/PCT
Perform the chlorhexidine bath and log-roll her to change her position	☐	☐	☐
Compare the pulses, color, warmth, sensation and movement of both feet and decide if the change is concerning	☐	☐	☐
Decide whether to give the 0900 beta blocker, lisinopril and furosemide after the plan is clarified	☐	☐	☐
Give the scheduled 1400 acetaminophen and gabapentin after the RN reviews her sedation score and respiratory rate	☐	☐	☐

5.5 Teaching Plan *(2 pts)*

List FOUR teaching topics this patient needs before and after surgery. Adapt each to her barriers (pain, anxiety, frailty, hearing and vision aids, spouse involvement) and say how you will check understanding.

Teaching topic	How you will teach it (adapted to barriers)	How you will confirm learning

SEALED PAGE. Read only after Parts 1 to 5 are complete.

Instructor addendum for the 0700-1500 shift. The EHR has no entries after 0700; use these events as if you had charted them.

0730-0900 Morning

- Glucose 187 at 0730; lispro correction given. Prednisone given 0800. The surgical team confirms the OR is tomorrow at 0800. They tell you: keep her NPO after midnight, give ONE beta blocker (carvedilol), hold lisinopril and furosemide, and the methotrexate will not be given.
- She states her last dose of apixaban was last night at 2000 (before the fall).

1215 The nerve block wears off

- Pain rises from 4/10 to 8/10 and she is restless. Morphine 2 mg IV given at 1220.

1300 Reassessment

- Pain 8/10. Right thigh looks larger and tense with new bruising spreading to the flank. R foot toes cool and pale, capillary refill 4 seconds, "pins and needles" (she has baseline neuropathy), pedal pulse faint. She is newly confused (CAM positive) and says "something is wrong."

Time	HR	BP	RR	SpO2	Temp	Pain	Other
0424	99	150/89	19	91% RA	98.3 F	6/10	baseline
1300	112	112/66	22	90% RA	98.1 F	8/10	glucose 241
1330	118	98/58	24	89% RA	98.4 F	9/10	toes cool; tense thigh; CAM positive
1340 labs							Hgb 7.6 (was 8.9); platelets 218; K 4.4; glucose 262

At 1400 her heparin 5,000 units SQ, gabapentin 300 mg and acetaminophen 1,000 mg are due.

PART 6: EVALUATION: WHAT HAPPENED AND WHAT NEXT?

Nursing process step: Evaluation | About 35 minutes | 8 points

Open the sealed page. It describes what happened during your shift (written by your instructor; it is not in the EHR). Do not change your Part 1-5 answers.

6.1 Evaluating Your Goals *(4 pts)*

For each goal you wrote in 5.1a and 5.1b, use data from the sealed page to say whether it was met, partly met, or not met, and what you will change.

Goal (copy from Part 5)	Data from the sealed page that you used	Met / Partly / Not	Revised plan

6.2 New Problems and Priority Actions *(2 pts)*

From the sealed page, name the TWO most urgent problems and for each the priority nursing actions (include who you notify and what you do about the medicines due at 1400).

6.3 Focused Nursing Note *(2 pts)*

Write a focused nursing note for the 1330 event (DAR, SOAP, or your school's format): data, action, and response, with the plan.

PART 7: CLINICAL JUDGMENT (NEXT-GENERATION NCLEX STYLE)

Nursing process step: Evaluation / Clinical judgment | About 30 minutes | 10 points

Answer by marking the boxes. These items use the same patient. Some scenarios are hypothetical and say so.

7.1 Select All That Apply *(2 pts)*

Which nursing actions are correct for this patient today, before surgery?

Select ALL that apply.

- ☐ Verify and document the time of her last home apixaban dose.
- ☐ Massage the right calf to prevent clots.
- ☐ Compare the color, warmth, pulses, sensation and toe movement of the right foot with the left, remembering her baseline foot neuropathy.
- ☐ Let her use the bedside commode with a walker.
- ☐ Give the lisinopril and furosemide at 0900 without checking because they are her home medicines.
- ☐ Question the lisinopril and furosemide with the provider before giving them.
- ☐ Continue her prednisone and ask whether stress-dose steroids are needed.
- ☐ Keep contact precautions and chlorhexidine bathing as ordered.

7.2 Expected or Report Now? *(2 pts)*

HYPOTHETICAL: she has the nail surgery tomorrow. On post-operative day 0 you find each of these. Mark one box per row.

	Expected after surgery	Report now
Incisional pain 4/10 that improves 30 minutes after oxycodone	☐	☐
R foot cool and pale, capillary refill 5 seconds, severe pain with passive toe movement	☐	☐
A 2 cm spot of pink drainage on the new dressing	☐	☐
HR 124, BP 88/50, new dizziness and a dressing saturated with blood	☐	☐

7.3 Bowtie *(2 pts)*

HYPOTHETICAL post-operative day 2: she suddenly has shortness of breath and sharp chest pain when she breathes in. HR 126, RR 30, SpO2 85% on room air, BP 112/70. Her right calf is swollen and tender.

Actions to Take (select 2)	Condition the patient is most likely experiencing (select 1)	Parameters to Monitor (select 2)

☐ Raise the head of the bed, apply oxygen, and call the rapid response team ☐ Walk her to the hallway to expand her lungs ☐ Massage the right calf ☐ Stay with her and apply continuous pulse oximetry and cardiac monitoring ☐ Give an extra IV fluid bolus	☐ Pulmonary embolism ☐ Surgical site infection ☐ Opioid overdose ☐ Hypoglycemia	☐ Oxygen saturation and respiratory effort ☐ Heart rate and rhythm (atrial fibrillation) ☐ Bowel sounds ☐ Incision drainage ☐ Urine color

7.4 Recognize Cues *(2 pts)*

Excerpt from a HYPOTHETICAL post-operative day 1 shift report. Which statements need follow-up NOW?

Draw a line under each statement that needs follow-up now. Then write the item numbers in the box below.

(1) Dressing clean, dry and intact; pain 3/10 with scheduled acetaminophen.

(2) She pulled out her IV at 0200 and keeps asking "who are these people?"; she was clear and oriented yesterday.

(3) Glucose 148 before breakfast.

(4) Last bowel movement was 4 days ago; she has taken oxycodone since surgery and has not had her laxative.

(5) Right foot warm with a palpable pulse and normal toe movement.

(6) She ate 75% of dinner.

(7) The red area over her sacrum no longer blanches when pressed.

(8) SpO2 93% on 2 L nasal cannula.

Item numbers that need follow-up:

7.5 Priority Actions *(2 pts)*

HYPOTHETICAL: on post-operative day 0 you find her pale and sweaty, BP 84/46, HR 128, with a dressing that is saturated with blood. Number the actions in the order you would do them.

Number the actions 1 (first) to 5 (last).

[___] Call the rapid response team or surgeon with SBAR.
[___] Apply direct pressure over the dressing, keep her flat, stay with her.
[___] Apply oxygen and attach monitors (SpO2 goal 88-92%).
[___] Confirm large-bore IV access in the LEFT arm, send type and cross and labs, prepare for blood.
[___] Hold the next heparin and blood pressure medicines and document.